

OWL Language

- A richer ontology language
- Relations between classes,
 - ▶ e.g., disjointness
- cardinality
 - ▶ e.g. “exactly one”
- Richer typing of properties
- Characteristics of properties (e.g., symmetry)

Tradeoff between Expressive Power and Efficient Reasoning Support

- The richer the language is, the more inefficient the reasoning support becomes
- Sometimes it crosses the border of noncomputability
- We need a compromise:
 - ▶ A language supported by reasonably efficient reasoners
 - ▶ A language that can express large classes of ontologies and knowledge

Examples of Reasoning Support

- Class membership

- ▶ If x is an instance of a class C , and C is a subclass of D , then we can infer that x is an instance of D

- Equivalence of classes

- ▶ If class A is equivalent to class B , and class B is equivalent to class C , then A is equivalent to C , too

Examples of Reasoning Support

- Consistency

- ▶ X instance of classes A and B , but A and B are disjoint
- ▶ This is an indication of an error in the ontology

- Classification

- ▶ Certain property-value pairs are a sufficient condition for membership in a class A ; if an individual x satisfies such conditions, we can conclude that x must be an instance of A

Uses for Reasoning

- Reasoning support is important for
 - ▶ checking the consistency of the ontology and the knowledge
 - ▶ checking for unintended relationships between classes
 - ▶ automatically classifying instances in classes
- Checks like the preceding ones are valuable for
 - ▶ designing large ontologies, where multiple authors are involved
 - ▶ integrating and sharing ontologies from various sources

Reasoning Support for OWL

- Semantics is a prerequisite for reasoning support
- Formal semantics and reasoning support are usually provided by
 - ▶ mapping an ontology language to a known logical formalism
 - ▶ using automated reasoners that already exist for those formalisms
- OWL is (partially) mapped on a description logic, and makes use of reasoners such as FaCT, RACER, Pellet
- Description logics are a subset of predicate logic for which efficient reasoning support is possible

<http://www.cs.man.ac.uk/~horrocks/FaCT/>

<http://www.racer-systems.com/> <http://pellet.owldl.com/>

<http://gaia.isti.cnr.it/~straccia>

OWL Language

- Three species of OWL
 - ▶ **OWL full** is union of OWL syntax and RDF (Undecidable)
 - ▶ **OWL DL** restricted to FOL fragment (decidable in NEXPTIME)
 - ▶ **OWL Lite** is “easier to implement” subset of OWL DL (decidable in EXPTIME)
- Semantic layering
 - ▶ OWL DL within **Description Logic (DL) fragment**
 - ▶ DL semantics **officially definitive**
- OWL DL based on *SHIQ* Description Logic
 - ▶ In fact it is equivalent to *SHOIN*(D_n) DL
- OWL DL Benefits from many years of DL research
 - ▶ Well defined **semantics**
 - ▶ **Formal properties** well understood (complexity, decidability)
 - ▶ Known **reasoning algorithms**
 - ▶ **Implemented systems** (highly optimised)

OWL Full

- It uses all the OWL languages primitives
- It allows the combination of these primitives in arbitrary ways with RDF and RDF Schema
- OWL Full is fully upward-compatible with RDF, both syntactically and semantically
- OWL Full is so powerful that it is undecidable
- No efficient reasoning support

- OWL DL (Description Logic) is a sublanguage of OWL Full that restricts application of the constructors from OWL and RDF
 - ▶ Application of OWL's constructors' to each other is disallowed
 - ▶ Therefore it corresponds to a well studied description logic
- OWL DL permits efficient reasoning support
- But we lose full compatibility with RDF:
 - ▶ Not every RDF document is a legal OWL DL document
 - ▶ Every legal OWL DL document is a legal RDF document.

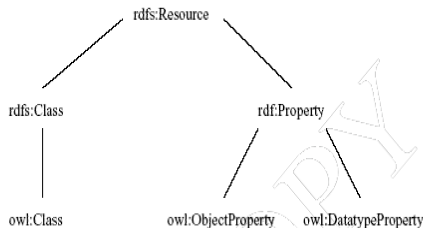
- An even further restriction limits OWL DL to a subset of the language constructors
 - ▶ E.g., OWL Lite excludes enumerated classes, disjointness statements, and arbitrary cardinality
- The advantage of this is a language that is easier to
 - ▶ grasp, for users
 - ▶ implement, for tool builders
- The disadvantage is restricted expressivity

Upward Compatibility between OWL Species

- Every legal OWL Lite ontology is a legal OWL DL ontology
- Every legal OWL DL ontology is a legal OWL Full ontology
- Every valid OWL Lite conclusion is a valid OWL DL conclusion
- Every valid OWL DL conclusion is a valid OWL Full conclusion

OWL Compatibility with RDF Schema

- All varieties of OWL use RDF for their syntax
- Instances are declared as in RDF, using RDF descriptions
- Typing information OWL constructors are specialisations of their RDF counterparts



OWL Syntactic Varieties

- OWL builds on RDF and uses RDF's XML-based syntax
- Other syntactic forms for OWL have also been defined:
 - ▶ An alternative, more readable XML-based syntax
 - ▶ A graphic syntax based on the conventions of UML
 - ▶ An abstract syntax, that is much more compact and readable than the XML languages
 - ▶ Description Logic syntax

From OWL to Description Logics

What Are Description Logics? (<http://dl.kr.org/>)

- A family of logic-based knowledge representation formalisms
 - ▶ Descendants of semantic networks and KL-ONE
 - ▶ Describe domain in terms of **concepts** (classes), **roles** (properties, relationships) and **individuals**
- Distinguished by:
 - ▶ **Formal semantics** (typically model theoretic)
 - ★ **Decidable fragments of FOL**
 - ★ Closely related to Propositional Modal & Dynamic Logics
 - ★ Closely related to Guarded Fragment of FOL, L2 and C2
 - ▶ Provision of **inference services**
 - ★ Decision procedures for key problems (e.g., satisfiability, subsumption)
 - ★ Implemented systems (highly optimised)

Short History of Description Logics

Phase 1: Mostly system development (early eighties)

- Incomplete systems (Back, Classic, Loom, Kander, ...)
- Based on **structural algorithms**

Phase 2: first tableaux algorithms and complexity results (mid-eighties - mid-nineties)

- Development of **tableaux algorithms** and **complexity results**
- Tableaux-based systems (Kris, Crack)
- Investigation of optimization techniques

Phase 3: Optimized systems for very expressive DLs (mid-nineties - ...)

- Tableaux algorithms for **very expressive** DLs
- **Highly optimised** tableau systems (FaCT, DLP, Racer)
- Relationship to modal logic and decidable fragments of FOL

Latest developments

Phase 4:

- Mature **implementations**
- Mainstream **applications** and Tools
 - ▶ Databases
 - ★ consistency of conceptual schema (EER, UML) using e.g. $\mathcal{DLR}$ (n -ary DL)
 - ★ schema integration
 - ★ Query subsumption (w.r.t. a conceptual schema)
 - ▶ Ontologies and the **Semantic Web** (and **Grid**)
 - ★ Ontology engineering (design, maintenance, integration)
 - ★ Reasoning with ontology-based markup (metadata)
 - ★ Service description and discovery
 - ▶ **Commercial** implemenations
 - ★ Cerebra, Racer

Description Logics Basics

- **Concept/Class**: names are equivalent to unary predicates
 - ▶ In general, concepts equiv to formulae with one free variable
- **Role or attribute**: names are equivalent to binary predicates
 - ▶ In general, roles equiv to formulae with two free variables
- **Taxonomy**: Concept and role hierarchies can be expressed
- **Individual**: names are equivalent to constants
- **Operators**: restricted so that:
 - ▶ Language is decidable and, if possible, of low complexity
 - ▶ No need for explicit use of variables
 - ★ Restricted form of $\exists$ and $\forall$
 - ▶ Features such as counting can be succinctly expressed

The DL Family

- A given DL is defined by set of concept and role forming operators
- Smallest propositionally closed DL is $\mathcal{ALC}$ (**A**ttributive **L**anguage with **C**omplement)
 - ▶ Concepts constructed using $\sqcap, \sqcup, \neg, \exists$ and $\forall$

Syntax			Example
$C, D \longrightarrow$	$\top$	(top concept)	
	$\perp$	(bottom concept)	
	A	(atomic concept)	Human
	$C \sqcap D$	(concept conjunction)	Human $\sqcap$ Male
	$C \sqcup D$	(concept disjunction)	Nice $\sqcap$ Rich
	$\neg C$	(concept negation)	$\neg$ Meat
	$\exists R.C$	(existential quantification)	$\exists$ hasChild.Blond
	$\forall R.C$	(universal quantification)	$\forall$ hasChild.Human

- **General inclusion axioms**: are of the form $C \sqsubseteq D$
 - ▶ “concept D subsumes concept C ”

HappyFather $\sqsubseteq$ *Man* $\sqcap \exists$ *hasChild.Female*

- ▶ **Concept definitions**: are of the form $A = C$ (equivalent to $A \sqsubseteq C, C \sqsubseteq A$)
- ▶ **Primitive concept definitions**: are of the form $A \sqsubseteq C$
- ▶ Sometimes, we have primitive and concept definitions only, where no atom can be defined more than once and no recursion is allowed
 - ★ in that case the concept definitions can be “unfolded”

- **Assertions**: are of the form $a:C$ and $(a, b):R$
 - ▶ “individual a is an instance of concept C ”
 - ▶ “individual a is related to b by means of R ”

John:Happy_Father
(John, Mary):has_child

- **(In)Equality Axioms**: are of the form $a = b$ and $a \neq b$

DL Semantics

- Semantics is given in terms of an **interpretation** $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$, where
 - ▶ $\Delta^{\mathcal{I}}$ is the **domain** (a non-empty set)
 - ▶ $\cdot^{\mathcal{I}}$ is an **interpretation function** that maps:
 - ★ **Concept** (class) name A into a subset $A^{\mathcal{I}}$ of $\Delta^{\mathcal{I}}$
 - ★ **Role** (property) name R into a subset $R^{\mathcal{I}}$ of $\Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$
 - ★ **Individual** name a into an element of $\Delta^{\mathcal{I}}$
 - ▶ Interpretation function $\cdot^{\mathcal{I}}$ is extended to concept expressions:

$$\begin{aligned}\top^{\mathcal{I}} &= \Delta^{\mathcal{I}} \\ \perp^{\mathcal{I}} &= \emptyset \\ (C_1 \sqcap C_2)^{\mathcal{I}} &= C_1^{\mathcal{I}} \cap C_2^{\mathcal{I}} \\ (C_1 \sqcup C_2)^{\mathcal{I}} &= C_1^{\mathcal{I}} \cup C_2^{\mathcal{I}} \\ (\neg C)^{\mathcal{I}} &= \Delta^{\mathcal{I}} \setminus C^{\mathcal{I}} \\ (\exists R.C)^{\mathcal{I}} &= \{x \in \Delta^{\mathcal{I}} \mid \exists y. \langle x, y \rangle \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\} \\ (\forall R.C)^{\mathcal{I}} &= \{x \in \Delta^{\mathcal{I}} \mid \forall y. \langle x, y \rangle \in R^{\mathcal{I}} \Rightarrow y \in C^{\mathcal{I}}\}\end{aligned}$$

- Example: assume $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ such that

$$\begin{aligned}\Delta^{\mathcal{I}} &= \{a, b, c, d, e, f, 1, 2, 4, 5, 8\} \\ \text{Person}^{\mathcal{I}} &= \{a, b, c, d\} \\ \text{Blonde}^{\mathcal{I}} &= \{b, d\} \\ \text{hasChild}^{\mathcal{I}} &= \{\langle a, b \rangle, \langle a, c \rangle, \langle c, d \rangle, \langle b, c \rangle\}\end{aligned}$$

- What is the interpretation of $\text{Person} \sqcap \exists \text{hasChild}.\text{Blond}$?

$$\begin{aligned}(\text{Person} \sqcap \exists \text{hasChild}.\text{Blond})^{\mathcal{I}} &= \text{Person}^{\mathcal{I}} \cap \{x \mid \exists y. \langle x, y \rangle \in \text{hasChild}^{\mathcal{I}} \wedge y \in \text{Blond}^{\mathcal{I}}\} \\ &= \{a, b, c, d\} \cap \{x \mid \exists y. \langle x, y \rangle \in \{\langle a, b \rangle, \langle a, c \rangle, \langle c, d \rangle, \langle b, c \rangle\} \wedge y \in \{b, d\}\} \\ &= \{a, b, c, d\} \cap \{a, c\} \\ &= \{a, c\}\end{aligned}$$

- Finally, we say that

- $\mathcal{I}$ is a model of $C \sqsubseteq D$, written $\mathcal{I} \models C \sqsubseteq D$, iff $C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$
- $\mathcal{I}$ is a model of $a:C$, written $\mathcal{I} \models a:C$, iff $a^{\mathcal{I}} \in C^{\mathcal{I}}$
- $\mathcal{I}$ is a model of $(a, b):R$, written $\mathcal{I} \models (a, b):R$, iff $\langle a^{\mathcal{I}}, b^{\mathcal{I}} \rangle \in R^{\mathcal{I}}$
- $\mathcal{I}$ is a model of $a = b$, written $\mathcal{I} \models a = b$, iff $a^{\mathcal{I}} = b^{\mathcal{I}}$
- $\mathcal{I}$ is a model of $a \neq b$, written $\mathcal{I} \models a \neq b$, iff $a^{\mathcal{I}} \neq b^{\mathcal{I}}$

Excerpt of **pizza** ontology ... (according to University of Manchester)

PizzaFruttiDiMare = *Pizza*
 $\sqcap \exists \text{hasTopping} . \text{MixedSeafoodTopping}$
 $\sqcap \exists \text{hasTopping} . \text{GarlicTopping}$
 $\sqcap \exists \text{hasTopping} . \text{TomatoTopping}$
 $\sqcap \forall \text{hasTopping} . (\text{MixedSeafoodTopping} \sqcup \text{GarlicTopping} \sqcup \text{TomatoTopping})$
 $\sqcap \exists \text{hasBase} . \text{PizzaBase}$

PizzaBase $\sqsupseteq$ *DeepPanBase* $\sqcup$ *ThinAndCrispyBase*

MixedSeafoodTopping $\sqsubseteq$ *FishTopping*

FishTopping $\sqsubseteq$ *PizzaTopping* $\sqcap \exists \text{hasSpiciness} . \text{Mild}$

disjoint(*FishTopping*, *MeatTopping*, *HerbSpiceTopping*)

functional(*hasSpiciness*)

Topping $\sqsubseteq$ $\forall \text{hasSpiciness} . (\text{Hot} \sqcup \text{Medium} \sqcup \text{Mild})$

Note: Simplification of acyclic ontologies

- Acyclic TBoxes can be unfolded or expanded
 - ▶ Replace each $A \sqsubseteq C$ with $A = C \sqcap A'$, where A' is a new atom
 - ▶ Given $A = C$, replace each occurrence of A with C
- Example:

<i>parent</i>	=	<i>person</i> $\sqcap$ $\exists$ <i>hasChild</i> . <i>person</i>
<i>woman</i>	=	<i>person</i> $\sqcap$ <i>femal</i>
<i>mother</i>	=	<i>parent</i> $\sqcap$ <i>femal</i>

Original TBox

<i>parent</i>	=	<i>person</i> $\sqcap$ $\exists$ <i>hasChild</i> . <i>person</i>
<i>woman</i>	=	<i>person</i> $\sqcap$ <i>femal</i>
<i>mother</i>	=	<i>person</i> $\sqcap$ $\exists$ <i>hasChild</i> . <i>person</i> $\sqcap$ <i>femal</i>

Unfolded TBox

DLs and First-Order-Logic

- $\mathcal{ALC}$ mapping to FOL: introduce
 - ▶ a unary predicate A for an atomic concept A
 - ▶ a binary predicate R for a role R
- Translate as follows

$t(\top, x)$	$=$	true	$t(\neg C, x)$	$=$	$\neg t(C, x)$
$t(\perp, x)$	$=$	false	$t(\exists R.C, x)$	$=$	$\exists y. R(x, y) \wedge t(C, y)$
$t(A, x) \mapsto$		$A(x)$	$t(\forall R.C, x)$	$=$	$\forall y. R(x, y) \Rightarrow t(C, y)$
$t(C_1 \sqcap C_2, x)$	$=$	$t(C_1, x) \wedge t(C_2, x)$	$t(C \sqsubseteq D)$	$=$	$\forall x. t(C, x) \Rightarrow t(D, x)$
$t(C_1 \sqcup C_2, x) \mapsto$		$t(C_1, x) \vee t(C_2, x)$	$t(a:C)$	$=$	$t(C, a)$
			$t((a, b):R)$	$=$	$R(a, b)$

- Example:

$$t(\text{HappyFather} \sqsubseteq \text{Man} \sqcap \exists \text{hasChild.Female}) = \\ \forall x. \text{HappyFather}(x) \Rightarrow (\text{Man}(x) \wedge (\exists y. \text{hasChild}(x, y) \wedge \text{Female}(y)))$$

$$t(a:\text{Man} \sqcap \exists \text{hasChild.Female}) = \\ \text{Man}(a) \wedge (\exists y. \text{hasChild}(a, y) \wedge \text{Female}(y))$$

Note on DL naming

$\mathcal{AL}$: $C, D \rightarrow \top \mid \perp \mid A \mid C \sqcap D \mid \neg A \mid \exists R. \top \mid \forall R. C$

$\mathcal{C}$: Concept negation, $\neg C$. Thus, $\mathcal{ALC} = \mathcal{AL} + \mathcal{C}$

$\mathcal{S}$: Used for $\mathcal{ALC}$ with transitive roles $\mathcal{R}_+$

$\mathcal{U}$: Concept disjunction, $C_1 \sqcup C_2$, e.g. *Male* $\sqcup$ *Female*

$\mathcal{E}$: Existential quantification, $\exists R. C$, e.g. $\exists \text{hasPart. Wheel}$

$\mathcal{H}$: Role inclusion axioms, $R_1 \sqsubseteq R_2$, e.g. *isComponentOf* $\sqsubseteq$ *isPartOf*

$\mathcal{N}$: Number restrictions, $(\geq n R)$ and $(\leq n R)$, e.g. $(\geq 3 \text{ has_Child})$ (has at least 3 children)

$\mathcal{Q}$: Qualified number restrictions, $(\geq n R. C)$ and $(\leq n R. C)$,
e.g. $(\leq 2 \text{ hasChild. Adult})$ (has at most 2 adult children)

$\mathcal{O}$: Nominals (singleton class), $\{a\}$, e.g. $\exists \text{hasChild. } \{mary\}$.

Note: $a:C$ equiv to $\{a\} \sqsubseteq C$ and $(a, b):R$ equiv to $\{a\} \sqsubseteq \exists R. \{b\}$

$\mathcal{I}$: Inverse role, R^- , e.g. *isPartOf* = *hasPart* $^-$

$\mathcal{F}$: Functional role, f , e.g. *fun*(*hasAge*)

$\mathcal{R}_+$: transitive role, e.g. *trans*(*isPartOf*)

For instance,

$$\begin{aligned} SHIF &= S + \mathcal{H} + \mathcal{I} + \mathcal{F} = \mathcal{ALCR}_+HIF \\ SHOIN &= S + \mathcal{H} + \mathcal{O} + \mathcal{I} + \mathcal{N} = \mathcal{ALCR}_+HOIN \end{aligned}$$

OWL-Lite (EXPTIME)

OWL-DL (NEXPTIME)



Semantics of additional constructs:

$\mathcal{H}$: Role inclusion axioms, $\mathcal{I} \models R_1 \sqsubseteq R_2$ iff $R_1^{\mathcal{I}} \subseteq R_2^{\mathcal{I}}$

$\mathcal{N}$: Number restrictions, $(\geq n R)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}}\}| \geq n\}$,
 $(\leq n R)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}}\}| \leq n\}$

$\mathcal{Q}$: Qualified number restrictions,
 $(\geq n R.C)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}| \geq n\}$,
 $(\leq n R.C)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}| \leq n\}$

$\mathcal{O}$: Nominals (singleton class), $\{a\}^{\mathcal{I}} = \{a^{\mathcal{I}}\}$

$\mathcal{I}$: Inverse role, $(R^{-})^{\mathcal{I}} = \{\langle x, y \rangle \mid \langle y, x \rangle \in R^{\mathcal{I}}\}$

$\mathcal{F}$: Functional role, $\mathcal{I} \models \text{fun}(f)$ iff $\forall x \forall y \forall z$ if $\langle x, y \rangle \in f^{\mathcal{I}}$ and $\langle x, z \rangle \in f^{\mathcal{I}}$ then $y = z$

$\mathcal{R}_+$: transitive role, $(R_+)^{\mathcal{I}} = \{\langle x, y \rangle \mid \exists z \text{ such that } \langle x, z \rangle \in R^{\mathcal{I}} \wedge \langle z, y \rangle \in R^{\mathcal{I}}\}$



Lisp-like Syntax of DLs formulae

In many systems the syntax is Lisp-like

$\top$	$\rightarrow$	<code>*top*</code>	$(\leq n R.C)$	$\rightarrow$	<code>(at-most n R C)</code>
$\perp$	$\rightarrow$	<code>*bottom*</code>	$\{a\}$	$\rightarrow$	<code>(one-of a)</code>
$\neg C$	$\rightarrow$	<code>(not C)</code>	R^-	$\rightarrow$	<code>(inv R)</code>
$C \sqcap D$	$\rightarrow$	<code>(and C D)</code>	$C \sqsubseteq D$	$\rightarrow$	<code>(implies C D)</code>
$C \sqcup D$	$\rightarrow$	<code>(or C D)</code>	$A = C$	$\rightarrow$	<code>(define-concept A C)</code>
$\exists R.C$	$\rightarrow$	<code>(some R C)</code>	$R \sqsubseteq P$	$\rightarrow$	<code>(implies-role R P)</code>
$\forall R.C$	$\rightarrow$	<code>(all R C)</code>	$\text{fun}(f)$	$\rightarrow$	<code>(functional f)</code>
$(\geq n R)$	$\rightarrow$	<code>(at-least n R)</code>	$\text{trans}(R)$	$\rightarrow$	<code>(transitive R)</code>
$(\leq n R)$	$\rightarrow$	<code>(at-most n R)</code>	$a:C$	$\rightarrow$	<code>(instance a C)</code>
$(\geq n R.C)$	$\rightarrow$	<code>(at-least n R C)</code>	$(a,b):R$	$\rightarrow$	<code>(related a b R)</code>

Example:

```
(define-concept Man (and Human Male))
(define-concept Happy_Father (and Man (some has_child Female)))

(instance john Happy_Father)
(related john mary has_child)
```

Concrete domains

- **Concrete domains:** reals, integers, strings, ...

(tim, 14):hasAge

(sf, "SoftComputing"):hasAcronym

(source1, "ComputerScience"):isAbout

(service2, "InformationRetrievalTool"):Matches

Minor = Person $\sqcap \exists hasAge. \leq_{18}$

- Semantics: a clean separation between “object” classes and concrete domains
 - ▶ $D = \langle \Delta_D, \Phi_D \rangle$
 - ▶ Δ_D is an interpretation domain
 - ▶ Φ_D is the set of concrete domain predicates d with a predefined arity n and **fixed** interpretation $d^D \subseteq \Delta_D^n$
 - ▶ Concrete properties: $R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta_D$
- Notation: (D) . E.g., $\mathcal{ALC}(D)$ is $\mathcal{ALC}$ + concrete domains

- Example: assume $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ such that

$$\begin{aligned}\Delta^{\mathcal{I}} &= \{a, b, c, d, e, f, 1, 2, 4, 5, 8\} \\ \text{Person}^{\mathcal{I}} &= \{a, b, c, d\}\end{aligned}$$

- Consider the following concrete domain with of some unary predicates ($n = 1$) over reals

- ▶ $\Delta_D = \mathbb{R}$,
- ▶ $\Phi_D = \{=m, \geq m, \leq m, > m, < m \mid m \in \mathbb{R}\}$
- ▶ the fixed interpretation of the predicates is

$$\begin{aligned}(=m)^D &= \{m\} & (>m)^D &= \{k \mid k > m\} \\ (\geq m)^D &= \{k \mid k \geq m\} & (<m)^D &= \{k \mid k < m\} \\ (\leq m)^D &= \{k \mid k \leq m\}\end{aligned}$$

- ▶ Concrete properties: $\text{hasAge}^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \mathbb{R}$

$$\text{hasAge}^{\mathcal{I}} = \{\langle a, 9 \rangle, \langle c, 20 \rangle, \langle b, 12 \rangle\}$$

- ▶ What is the interpretation of $\text{Person} \sqcap \exists \text{hasAge}. \leq_{18}$?

$$\begin{aligned}(\text{Person} \sqcap \exists \text{hasAge}. \leq_{18})^{\mathcal{I}} &= \text{Person}^{\mathcal{I}} \cap \{x \mid \exists y \in \mathbb{R} \text{ such that } \langle x, y \rangle \in \text{hasAge}^{\mathcal{I}} \wedge y \in (\leq_{18})^{\mathcal{I}}\} \\ &= \{a, b, c, d\} \cap \{x \mid \exists y. \langle x, y \rangle \in \{\langle a, 9 \rangle, \langle c, 20 \rangle, \langle b, 12 \rangle\} \wedge y \leq 18\} \\ &= \{a, b, c, d\} \cap \{a, b\} \\ &= \{a, b\}\end{aligned}$$



DL Knowledge Base

- A DL **Knowledge Base** is a pair $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$, where
 - ▶ $\mathcal{T}$ is a **TBox**
 - ★ containing general inclusion axioms of the form $C \sqsubseteq D$,
 - ★ concept definitions of the form $A = C$
 - ★ primitive concept definitions of the form $A \sqsubseteq C$
 - ★ role inclusions of the form $R \sqsubseteq P$
 - ★ role equivalence of the form $R = P$
 - ▶ $\mathcal{A}$ is a **ABox**
 - ★ containing assertions of the form $a:C$
 - ★ containing assertions of the form $(a, b):R$
 - ★ containing (in) equality Axioms of the form $a = b$ and $a \neq b$
- An interpretation $\mathcal{I}$ is a model of $\mathcal{K}$, written $\mathcal{I} \models \mathcal{K}$ iff $\mathcal{I} \models \mathcal{T}$ and $\mathcal{I} \models \mathcal{A}$, where
 - ▶ $\mathcal{I} \models \mathcal{T}$ ($\mathcal{I}$ is a model of $\mathcal{T}$) iff $\mathcal{I}$ is a model of each element in $\mathcal{T}$
 - ▶ $\mathcal{I} \models \mathcal{A}$ ($\mathcal{I}$ is a model of $\mathcal{A}$) iff $\mathcal{I}$ is a model of each element in $\mathcal{A}$

OWL DL as Description Logic

Concept/Class constructors:

Abstract Syntax	DL Syntax	Example
Descriptions (C)		
A (URI reference) owl:Thing owl:Nothing	A $\top$ $\perp$	Conference
intersectionOf($C_1 C_2 \dots$) unionOf($C_1 C_2 \dots$) complementOf(C) oneOf($a_1 \dots$)	$C_1 \sqcap C_2$ $C_1 \sqcup C_2$ $\neg C$ $\{a_1, \dots\}$	Reference $\sqcap$ Journal Organization $\sqcup$ Institution $\neg$ MasterThesis $\{\text{"WISE"}, \text{"ISWC"}, \dots\}$
restriction(R someValuesFrom(C)) restriction(R allValuesFrom(C)) restriction(R hasValue(o)) restriction(R minCardinality(n)) restriction(R maxCardinality(n))	$\exists R.C$ $\forall R.C$ $\exists R.\{o\}$ $(\geq n R)$ $(\leq n R)$	$\exists$ parts.InCollection $\forall$ date.Date $\exists$ date.{2005} $(\geq 1 \text{ location})$ $(\leq 1 \text{ publisher})$
restriction(U someValuesFrom(D)) restriction(U allValuesFrom(D)) restriction(U hasValue(v)) restriction(U minCardinality(n)) restriction(U maxCardinality(n))	$\exists U.D$ $\forall U.D$ $\exists U. =v\}$ $(\geq n U)$ $(\leq n U)$	$\exists$ issue.integer $\forall$ name.string $\exists$ series.="LNCS" $(\geq 1 \text{ title})$ $(\leq 1 \text{ author})$

Note: R is an abstract role, while U is a concrete property of arity two.



Axioms:

Abstract Syntax	DL Syntax	Example
Axioms Class(<i>A</i> partial $C_1 \dots C_n$) Class(<i>A</i> complete $C_1 \dots C_n$) EnumeratedClass(<i>A</i> $o_1 \dots o_n$) SubClassOf($C_1 C_2$) EquivalentClasses($C_1 \dots C_n$) DisjointClasses($C_1 \dots C_n$)	$A \sqsubseteq C_1 \sqcap \dots \sqcap C_n$ $A = C_1 \sqcap \dots \sqcap C_n$ $A = \{o_1\} \sqcup \dots \sqcup \{o_n\}$ $C_1 \sqsubseteq C_2$ $C_1 = \dots = C_n$ $C_i \sqcap C_j = \perp, i \neq j$	$Human \sqsubseteq Animal \sqcap Biped$ $Man = Human \sqcap Male$ $RGB = \{r\} \sqcup \{g\} \sqcup \{b\}$ $Male \sqcap Female \sqsubseteq \perp$
ObjectProperty(<i>R</i> super ($R_1 \dots super (R_n)$) domain($C_1 \dots domain(C_n)$) range($C_1 \dots range(C_n)$) [inverseof(<i>P</i>)] [symmetric] [functional] [Inversefunctional] [Transitive]) SubPropertyOf($R_1 R_2$) EquivalentProperties($R_1 \dots R_n$) AnnotationProperty(<i>S</i>)	$R \sqsubseteq R_i$ $(\geq 1 R) \sqsubseteq C_i$ $\top \sqsubseteq \forall R. D_i$ $R = P^-$ $R = R^-$ $\top \sqsubseteq (\leq 1 R)$ $\top \sqsubseteq (\leq 1 R^-)$ $Tr(R)$ $R_1 \sqsubseteq R_2$ $R_1 = \dots = R_n$	$HasDaughter \sqsubseteq hasChild$ $(\geq 1 hasChild) \sqsubseteq Human$ $\top \sqsubseteq \forall hasChild. Human$ $hasChild = hasParent^-$ $similar = similar^-$ $\top \sqsubseteq (\leq 1 hasMother)$ $Tr(ancestor)$ $cost = price$

Abstract Syntax	DL Syntax	Example
DatatypeProperty(U super (U_1)... super (U_n) domain(C_1)...domain(C_n) range(D_1)...range(D_n) [functional]) SubPropertyOf($U_1 U_2$) EquivalentProperties($U_1 \dots U_n$)	$U \sqsubseteq U_i$ $(\geq 1 U) \sqsubseteq C_i$ $\top \sqsubseteq \forall U.D_i$ $\top \sqsubseteq (\leq 1 U)$ $U_1 \sqsubseteq U_2$ $U_1 = \dots = U_n$	$(\geq 1 \text{ hasAge}) \sqsubseteq \text{Human}$ $\top \sqsubseteq \forall \text{ hasAge.posInteger}$ $\top \sqsubseteq (\leq 1 \text{ hasAge})$ $\text{hasName} \sqsubseteq \text{hasFirstName}$
Individuals		
Individual(o type (C_1)... type (C_n) value($R_1 o_1$)...value($R_n o_n$) value($U_1 v_1$)...value($U_n v_n$) SameIndividual($o_1 \dots o_n$) DifferentIndividuals($o_1 \dots o_n$)	$o:C_i$ $(o, o_i):R_i$ $(o, v_i):U_i$ $o_1 = \dots = o_n$ $o_i \neq o_j, i \neq j$	$\text{tim}:\text{Human}$ $(\text{tim}, \text{mary}):\text{hasChild}$ $(\text{tim}, 14):\text{hasAge}$ $\text{president_Bush} = \text{G.W.Bush}$ $\text{john} \neq \text{peter}$
Symbols		
Object Property R (URI reference) Datatype Property U (URI reference) Individual o (URI reference) Data Value v (RDF literal)	R U U U	hasChild hasAge tim $\text{"International Conference on Semantic Web"}$

XML representation of OWL statements

Example:

Person $\sqcap \forall hasChild. (Doctor \sqcup \exists hasChild. Doctor)$

```
intersectionOf(  
  Person  
  restriction(hasChild  
    allValuesFrom(unionOf(Doctor  
      restriction(hasChild  
        someValuesFrom(Doctor)))
```

```
<owl:Class>  
  <owl:intersectionOf rdf:parseType=" collection">  
    <owl:Class rdf:about="#Person"/>  
    <owl:Restriction>  
      <owl:onProperty rdf:resource="#hasChild"/>  
      <owl:allValuesFrom>  
        <owl:unionOf rdf:parseType=" collection">  
          <owl:Class rdf:about="#Doctor"/>  
          <owl:Restriction>  
            <owl:onProperty rdf:resource="#hasChild"/>  
            <owl:someValuesFrom rdf:resource="#Doctor"/>  
          </owl:Restriction>  
        </owl:unionOf>  
      </owl:allValuesFrom>  
    </owl:Restriction>  
  </owl:intersectionOf>  
</owl:Class>
```

OWL Lite as Description Logic

- Prohibits union and complements
- Restricts intersection to implicit intersections in the frame-like class axioms
- Limits all embedded descriptions to concept names
- Does not allow individuals to show up in description or class axioms
- Limits role filler cardinalities to 0 or 1

Concept/Class constructors:

Abstract Syntax	DL Syntax	Example
Descriptions (C) A (URI reference) <code>owl:Thing</code> <code>owl:Nothing</code>	A $\top$ $\perp$	Conference
<code>restriction(R someValuesFrom(C))</code> <code>restriction(R allValuesFrom(C))</code> <code>restriction(R minCardinality(n))</code> <code>restriction(R maxCardinality(n))</code>	$\exists R.C$ $\forall R.C$ $(\geq n R), n \in \{0, 1\}$ $(\leq n R), n \in \{0, 1\}$	<code>$\exists$parts.InCollection</code> <code>$\forall$date.Date</code> <code>$(\geq 1$ location)</code> <code>$(\leq 1$ publisher)</code>
<code>restriction(U someValuesFrom(D))</code> <code>restriction(U allValuesFrom(D))</code> <code>restriction(U minCardinality(n))</code> <code>restriction(U maxCardinality(n))</code>	$\exists U.D$ $\forall U.D$ $(\geq n U), n \in \{0, 1\}$ $(\leq n U), n \in \{0, 1\}$	<code>$\exists$issue.integer</code> <code>$\forall$name.string</code> <code>$(\geq 1$ title)</code> <code>$(\leq 1$ author)</code>

Note: R is an abstract role, while U is a concrete property of arity two.

Axioms:

Abstract Syntax	DL Syntax	Example
Class(<i>A</i> partial $C_1 \dots C_n$) Class(<i>A</i> complete $C_1 \dots C_n$) EquivalentClasses($C_1 \dots C_n$) ObjectProperty(<i>R</i> super (R_1)... super (R_n) domain(C_1)...domain(C_n) range(C_1)...range(C_n) [inverseof(<i>P</i>)] [symmetric] [functional] [Inversefunctional] [Transitive]) SubPropertyOf($R_1 R_2$) EquivalentProperties($R_1 \dots R_n$) AnnotationProperty(<i>S</i>)	$A \sqsubseteq C_1 \sqcap \dots \sqcap C_n$ $A = C_1 \sqcap \dots \sqcap C_n$ $C_1 = \dots = C_n$ $R \sqsubseteq R_i$ $(\geq 1 R) \sqsubseteq C_i$ $\top \sqsubseteq \forall R.D_i$ $R = P^-$ $R = R^-$ $\top \sqsubseteq (\leq 1 R)$ $\top \sqsubseteq (\leq 1 R^-)$ $Tr(R)$ $R_1 \sqsubseteq R_2$ $R_1 = \dots = R_n$	$Human \sqsubseteq Animal \sqcap Biped$ $Man = Human \sqcap Male$ $HasDaughter \sqsubseteq hasChild$ $(\geq 1 hasChild) \sqsubseteq Human$ $\top \sqsubseteq \forall hasChild.Human$ $hasChild = hasParent^-$ $similar = similar^-$ $\top \sqsubseteq (\leq 1 hasMother)$ $Tr(ancestor)$ $cost = price$

Abstract Syntax	DL Syntax	Example
DatatypeProperty(U super (U_1)... super (U_n) domain(C_1)...domain(C_n) range(D_1)...range(D_n) [functional]) SubPropertyOf($U_1 U_2$) EquivalentProperties($U_1 \dots U_n$)	$U \sqsubseteq U_i$ $(\geq 1 U) \sqsubseteq C_i$ $\top \sqsubseteq \forall U.D_i$ $\top \sqsubseteq (\leq 1 U)$ $U_1 \sqsubseteq U_2$ $U_1 = \dots = U_n$	$(\geq 1 \text{ hasAge}) \sqsubseteq \text{Human}$ $\top \sqsubseteq \forall \text{ hasAge.posInteger}$ $\top \sqsubseteq (\leq 1 \text{ hasAge})$ $\text{hasName} \sqsubseteq \text{hasFirstName}$
Individuals		
Individual(o type (C_1)... type (C_n) value($R_1 o_1$)...value($R_n o_n$) value($U_1 v_1$)...value($U_n v_n$) SameIndividual($o_1 \dots o_n$) DifferentIndividuals($o_1 \dots o_n$)	$o:C_i$ $(o, o_i):R_i$ $(o, v_i):U_i$ $o_1 = \dots = o_n$ $o_i \neq o_j, i \neq j$	$\text{tim}:\text{Human}$ $(\text{tim}, \text{mary}):\text{hasChild}$ $(\text{tim}, 14):\text{hasAge}$ $\text{president_Bush} = \text{G.W.Bush}$ $\text{john} \neq \text{peter}$
Symbols		
Object Property R (URI reference) Datatype Property U (URI reference) Individual o (URI reference) Data Value v (RDF literal)	R U U U	hasChild hasAge tim $\text{"International Conference on Semantic Web"}$